



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - FEBRUARY/MARCH 2023

SUBJECT: ECE 1071 / ECE-1071 - BASIC ELECTRONICS

(MAKEUP)

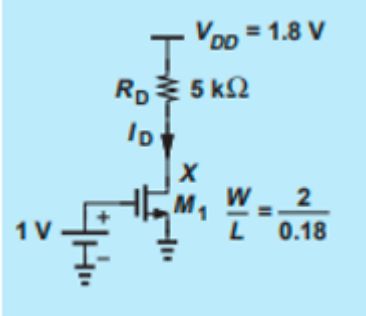
BASIC ELECTRONICS [ECE 1071]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) A silicon diode has a reverse saturation current of 12nA at 20°C .
i) Find diode current when it is forward biased by 0.65V . (4)
ii) Find the dynamic resistance offered by the diode at 20°C .
- 1B) Draw V-I characteristics of the 'Si' diode. Explain the different types of breakdowns that can occur in a PN junction diode. (3)
- 1C) Calculate the bias current of M_1 shown in the figure below. Assume $\mu_n C_{ox} = 90 \mu\text{A/V}^2$ and $V_{TH} = 0.5 \text{ V}$. If the gate voltage increases by 20 mV in Q_1 , what is the change in the drain voltage? (3)
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- 2A) Draw the Zener voltage regulator circuit. A Zener diode has a breakdown voltage of 10V . It is supplied from a voltage source varying between $20\text{V} - 40\text{V}$ in series with a resistance of 820Ω and load resistance of $1\text{k}\Omega$. Obtain the minimum and maximum Zener currents. (4)
- 2B) An AC supply of 230V , 50Hz is applied to a half wave rectifier circuit through a transformer with turns ratio $10:1$. Determine (a) the output DC and RMS voltage (b) Efficiency and ripple factor. Given load resistance of $1\text{k}\Omega$ connected across the circuit. (3)
- 2C) With a neat circuit diagram, explain the working of a center-tapped full wave rectifier. Draw the corresponding input and output waveforms. (3)
- 3A) A combinational logic circuit has four inputs A, B, C, and D, and one output F. The output F will be high only when two or more inputs are high and F will be zero otherwise. Obtain the truth table for F. Use Karnaugh map to find simplified SOP and implement the same using basic logic gates. (4)
- 3B) Derive an expression for the output of op-amp summing amplifier circuit with V_1 and V_2 as the inputs. Determine the resistor values to get an output $V_O = -(5V_2 + 3V_1)$ for this circuit. Assume $R_F = 10\text{k}\Omega$. (3)

- 3C) Derive an expression for inverting op-amp differentiator. Determine the output voltage for a sinewave input signal of 1V peak and draw its output waveform. Assume $C=0.1\mu\text{F}$ and $R=1.5\text{k}\Omega$. (3)
- 4A) Simplify the following SOP expression using K-Map and implement using basic gates only. (4)
 $F(A, B, C, D) = \sum m(1, 4, 7, 10, 13)$
- 4B) Realise a 3-bit down counter using the negative edge-triggered JK flip-flops. Draw the timing diagram for the same. (3)
- 4C) Draw the circuit diagram of Serial-in Serial-out 4-bit shift register. Also, show the shifting of the data 1011 to the right using the data transfer table. (3)
- 5A) Define Amplitude modulation and sketch the amplitude modulated wave for a sine wave modulating signal. A Certain AM transmitter radiates 9kW of power with carrier unmodulated and 10.125kW of power when the carrier is sinusoidally modulated. Calculate the modulation index. (4)
- 5B) For the binary data 10110, sketch the ASK, FSK, and PSK signals. (3)
- 5C) Draw the block diagram of the Communication System and explain the function of each block. (3)

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